

Batch:	Roll No.:
Experiment / assignment / tutorial No. 08	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

Title: AM Radio Receiver in Simulink

Objectives:

1. To understand basic network design in Simulink
2. To design and implement AM radio receiver in Simulink

OUTCOME:

1. understand working of analog communication transmitter and receiver systems
2. understand noise effect on the performance of communication systems.

1) Getting started with Simulink:

- To get started with Simulink, create a simple network of amplifying a sine wave by some factor and plot original and amplified signal on the scope.
- For the above task, copy network, necessary parameter settings and oscilloscope view.

2) AM Receiver design using Simulink

- **Open the example:** <https://in.mathworks.com/help/sps/ug/am-radio-receiver.html>
- Understand line by line working of radio receiver.
- Open the model in Simulink
- Take screenshot of each block and write small description below it and its significance.
- Take final oscilloscope signal
- Write conclusion
- **Post lab activity:**
 - What is the main purpose of intermediate frequency (IF) stage in a super heterodyne receiver? (Justify the answer)- 2 marks
 - To convert the signal to a lower frequency
 - To facilitate easier and more selective filtering
 - To amplify the received signal
 - To demodulate the signal

Somaiya Vidyavihar University
K J Somaiya School of Engineering

- A super heterodyne receiver with an IF of 450 KHz is tuned to a signal at 1200 KHz. The image frequency is_____ (show calculation)- 2 marks
 - 750 KHz
 - 900 KHz
 - 2100 KHz
 - 1650 KHz

- What is significance of selectivity and sensitivity in radio receiver? (write significance and not definition)- 1 mark

Signature of faculty in-charge